

$$F(x, y) = 3x^5y^3 + 10x^4y^2 + 2xy^4 - 7x + 4y + 185$$

$$F'_x = 15x^4y^3 + 40x^3y^2 + 2y^4 - 7$$

$$(a^x)' = a^x \ln a \cdot x'$$

$$F'_y = 9x^5y^2 + 20x^4y + 8xy^3 + 4$$

$$F(x, y) = (\sin 3x)^{\cos 2y^2}$$

$$F'_x = \cos 2y^2 \cdot (\sin 3x)^{\cos 2y^2 - 1} \cdot \cos 3x \cdot 3$$

$$F'_y = (\sin 3x)^{\cos 2y^2} \cdot \ln(\sin 3x) \cdot (-\sin 2y^2) \cdot 4y$$

Oblicz pochodne cząstkowe F do drugiego rzędu w punkcie

$$F(x, y) = x^{\lg y}$$

$$F'_{xy} = \frac{1}{\cos^2 y} \cdot x^{\lg y - 1} + \lg y \cdot x^{\lg y - 1} \cdot \ln x \cdot \frac{1}{\cos^2 y}$$

$$F'_x = \lg y \cdot x^{\lg y - 1}$$

$$F'_{xy} = \frac{x^{\lg y - 1}}{\cos^2 y} (1 + \lg y \ln x) =$$

$$F'_y = x^{\lg y} \ln x \cdot \frac{1}{\cos^2 y}$$

$$F'_{xx} = \lg y (\lg y - 1) \cdot x^{\lg y - 2} = F'_{yx}$$

$$F_{xy}' = \ln x (x^{\operatorname{tg} y} \ln x \cdot \frac{1}{\cos^2 y} \cdot \frac{1}{\cos^2 y} + x^{\operatorname{tg} y} \cdot \frac{\sin 2y}{\cos^4 y})$$

Znaleźć ekstremum funkcji (jeśli istnieje) typowe na egzamin

$$F(x, y) = x^2 + xy + y^2 - 2x - y \quad \underline{D = \mathbb{R}^2}$$

Szukać punktów stacjonarnych $F_x'(x_0, y_0) = 0$ i $F_y'(x_0, y_0) = 0$

$$F_x' = 2x + y - 2$$

$$F_y' = 2y + x - 1$$

$$\underline{(1, 0)}$$

$$\begin{cases} 2x + y = 2 \\ 2y + x = 1 \quad \parallel \cdot (-2) \end{cases}$$

$$\begin{cases} 2x + y = 2 \\ -4y - 2x = -2 \end{cases}$$

$$-3y = 0$$

$$y = 0$$

$$x = 1$$

$$F_{xx}'' = 2$$

$$F_{xy}'' = F_{yx}'' = 1$$

$$F_{yy}'' = 2$$

$$H_{(1,0)} = \begin{vmatrix} \boxed{2} & 1 \\ 1 & 2 \end{vmatrix} > 0$$

↓
MIN lokale

$$F(x, y) = x^2 + xy + y^2 - 4 \ln|x| - 10 \ln|y|$$

$$D: x \neq 0 \quad y \neq 0 \Rightarrow \mathbb{R}^2 \setminus \{0x, 0y\}$$

$$F'_x = 2x + y - \frac{4}{x}$$

$$F'_y = 2y + x - \frac{10}{y}$$

$$F_{xx}'' = 2 + \frac{4}{x^2}$$

$$F_{xy}'' = 1 = F_{yx}''$$

$$F_{yy}'' = 2 + \frac{10}{y^2}$$

$$F'_x(x_0, y_0) = 0 \wedge F'_y(x_0, y_0) = 0$$

$$\begin{cases} 2x + y - \frac{4}{x} = 0 \\ 2y + x - \frac{10}{y} = 0 \end{cases}$$

$$y = \frac{4}{x} - 2x = \frac{4-2x^2}{x}$$

$$361 - 192 = 169$$

$$\frac{8-4x^2}{x} + x - 5 \frac{x}{2-x^2} = 0 \parallel \cdot x(2-x^2)$$

$$(8-4x^2)(2-x^2) + x^2(2-x^2) - 5x^2 = 0$$

$$16 - 8x^2 - 8x^2 + 4x^4 + 2x^2 - x^4 - 5x^2 = 0$$

$$3x^4 - 19x^2 + 16 = 0$$

$$3t^2 - 19t + 16 = 0$$

$$t = x^2 \quad t \geq 0$$

$$\Delta = 169 \quad \sqrt{\Delta} = 13$$

$$t_1 = \frac{19-13}{6} = 1 \quad t_2 = \frac{19+13}{6} = \frac{16}{3}$$

$$y = \frac{4}{x} - 2x$$

$$x = 1 \vee x = -1 \vee x = \frac{4+\sqrt{3}}{3} \vee x = -\frac{4+\sqrt{3}}{3}$$

$$y = 2 \vee y = -2 \vee y = \sqrt{3} - \frac{8+\sqrt{3}}{3} \vee y = -\sqrt{3} + \frac{8+\sqrt{3}}{3}$$

$$y = -\frac{5+\sqrt{3}}{3} \vee y = \frac{5+\sqrt{3}}{3}$$

$$(1, 2) \vee (-1, -2) \vee \left(\frac{4+\sqrt{3}}{3}, -\frac{5+\sqrt{3}}{3}\right) \vee \left(-\frac{4+\sqrt{3}}{3}, \frac{5+\sqrt{3}}{3}\right)$$

$$H_{(1,2)} = \begin{vmatrix} \textcircled{6} & 1 \\ 1 & 2 + \frac{10}{4} \end{vmatrix} > 0$$

↓
min lokale

$$H_{(-1,-2)} = \begin{vmatrix} \textcircled{6} & 1 \\ 1 & 2 + \frac{10}{4} \end{vmatrix} > 0$$

↓
min lokale

$$H_{\left(\frac{4+\sqrt{3}}{3}, -\frac{5+\sqrt{3}}{3}\right)} = \begin{vmatrix} \cos' > 2 & 1 \\ 1 & \cos' > 2 \end{vmatrix}$$

↓
min lokale

→ min lokale

$$F(x, y) = x^2 + xy + y^2 + \frac{8}{x} + \frac{8}{y}$$

$$D: x \neq 0, y \neq 0$$

$$F'_x = 2x + y - \frac{8}{x^2}$$

$$F'_y = 2y + x - \frac{8}{y^2}$$

$$F''_{xx} = 2 + \frac{16}{x^3}$$

$$F''_{xy} = 1 = F''_{yx}$$

$$F''_{yy} = 2 + \frac{16}{y^3}$$

$$\begin{cases} 2x + y - \frac{8}{x^2} = 0 \\ 2y + x - \frac{8}{y^2} = 0 \end{cases}$$

Równania
symetryczne,
więc $x = y$

$$3x - \frac{8}{x^2} = 0 \parallel \cdot x^2$$

$$3x^3 - 8 = 0$$

$$x^3 = \frac{8}{3}$$

$$x = \frac{2}{\sqrt[3]{3}} \quad \frac{\sqrt[3]{9}}{3}$$

$$y = \frac{\sqrt[3]{9}}{3}$$

$$H_{\left(\frac{2}{\sqrt[3]{3}}, \frac{\sqrt[3]{9}}{3}\right)} = \begin{vmatrix} \textcircled{8} & 1 \\ 1 & 8 \end{vmatrix} > 0$$

↙
min lokalne

$$F(x, y) = e^{2x+3y}(8x^2 - 6xy + 3y^2)$$

$$F'_x = 2e^{2x+3y}(8x^2 - 6xy + 3y^2) + e^{2x+3y}(16x - 6y) =$$

$$e^{2x+3y}(16x^2 - 12xy + 6y^2 + 16x - 6y)$$

$$F'_y = 3e^{2x+3y}(8x^2 - 6xy + 3y^2) + e^{2x+3y}(6y - 6x) =$$

$$= e^{2x+3y}(24x^2 - 18xy + 9y^2 + 6y - 6x) \quad (0,0) \vee (-\frac{1}{4}, -\frac{1}{2})$$

$$8x^2 - 6xy + 3y^2 + 8x - 3y = 0$$

$$8x^2 - 6xy + 3y^2 + 2y - 2x = 0$$

$$8x - 3y = 2y - 2x$$

$$10x = 5y$$

$$y = 2x$$

$$8x^2 - 12x^2 + 12x^2 + 8x - 6x = 0$$

$$8x^2 + 2x = 0$$

$$2x(4x+1) = 0$$

$$x = 0 \quad \vee \quad x = -\frac{1}{4}$$

$$y = 0 \quad \vee \quad y = -\frac{1}{2}$$

$$F''_{xx} = 2e^{2x+3y}(16x^2 - 12xy + 6y^2 + 16x - 6y) + e^{2x+3y}(32x - 12y + 16)$$

$$F''_{xy} = 3e^{2x+3y}(16x^2 - 12xy + 6y^2 + 16x - 6y) + e^{2x+3y}(-12x + 12y - 6)$$

$$F''_{yx} = 3e^{2x+3y}(16x^2 - 12xy + 6y^2 + 16x - 6y) + e^{2x+3y}(-12x + 12y - 6)$$

$$F''_{yy} = 3e^{2x+3y}(24x^2 - 18xy + 9y^2 + 16y - 6x) + e^{2x+3y}(-18x + 18y + 6)$$

$$H_{(0,0)} = \begin{vmatrix} \textcircled{16} & -6 \\ -6 & 6 \end{vmatrix} > 0$$

↙ min lokalne

$$H_{(-\frac{1}{4}, -\frac{1}{2})} = \begin{vmatrix} e^{-2}(-8 + 6 + 16) & e^{-2}(\overbrace{3-6-6}^{-9}) \\ e^{-2}(-9) & e^{-2}(\frac{18}{4} - 9 + 6) \end{vmatrix} =$$

$$= \begin{vmatrix} 14e^{-2} & -9e^{-2} \\ -9e^{-2} & \frac{3}{2}e^{-2} \end{vmatrix} < 0$$

BRAK EKSTREMUM